

Bipolar disorder, or manic-depressive disorder, involves extreme swings of mood and affects about 1% of the world's population. The manic phase is characterized by high self-esteem, increased energy, a flow of ideas, overtalkativeness, and increased risk taking. In its milder forms, this phase is sometimes associated with great creativity, and some well-known artists, musicians, and literary figures (Vincent van Gogh, Robert Schumann, Virginia Woolf, and Ernest Hemingway, to name a few) have had productive periods during manic phases. The depressive phase comes with lessened motivation, sense of worth, and ability to feel pleasure, as well as sleep disturbances. These symptoms can be so severe that affected individuals attempt suicide.

Major depressive and bipolar disorders are among the nervous system disorders for which therapies are available. Many drugs used to treat depressive illness, including fluoxetine (Prozac), increase activity of biogenic amines in the brain.

The Brain's Reward System and Drug Addiction

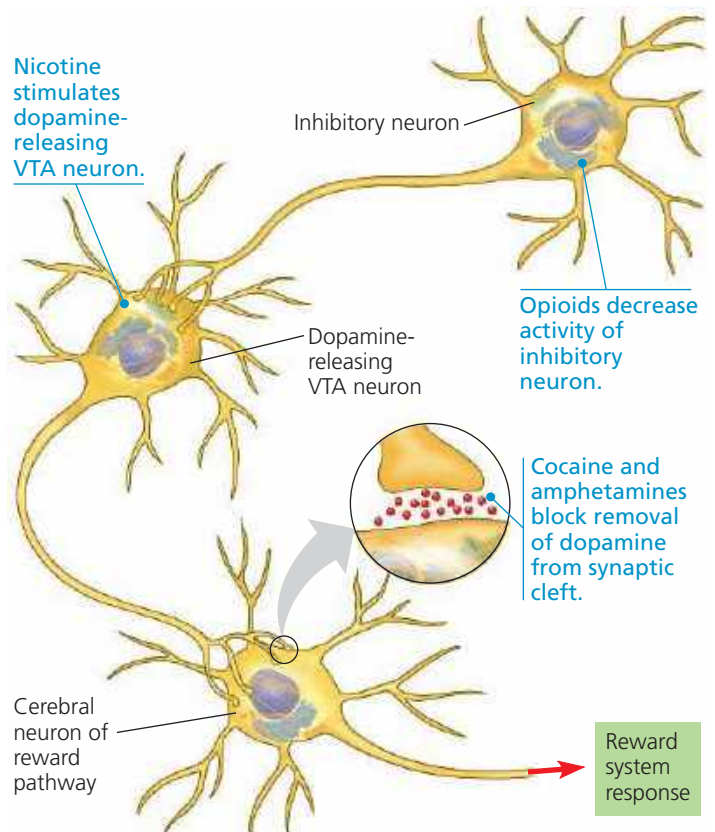
Emotions are strongly influenced by a neuronal circuit in the brain called the *reward system*. The reward system provides motivation for activities that enhance survival and reproduction, such as eating in response to hunger, drinking when thirsty, and engaging in sexual activity when aroused. Inputs to the reward system are received by neurons in the *ventral tegmental area (VTA)*, a region located within the midbrain (see Figure 49.11). When activated, these neurons release the neurotransmitter dopamine from their synaptic terminals (**Figure 49.24**). Targets of this dopamine signaling include the nucleus accumbens and the prefrontal cortex.

The brain's reward system is dramatically affected by drug addiction, a disorder characterized by compulsive consumption of a drug and loss of control in limiting intake. Addictive drugs range from sedatives to stimulants and include alcohol, cocaine, and nicotine, as well as opioids, such as heroin, fentanyl and oxycodone. All enhance the activity of the dopamine pathway (see Figure 49.24). As addiction develops, there are also long-lasting changes in the reward circuitry. The result is a craving for the drug independent of any pleasure associated with consumption. In 2018, the Centers for Disease Control and Prevention reported that an average of 130 Americans died from opioid overdose every day.

Laboratory animals are highly valuable in modeling and studying addiction. Rats, for example, will provide themselves with heroin, cocaine, or amphetamine when given a dispensing system linked to a lever in their cage. Furthermore, they exhibit addictive behavior, continuing to self-administer the drug rather than seek food, even to the point of starvation.

As scientists expand their knowledge about the brain's reward system and the various forms of addiction, there is hope that the insights will lead to more effective prevention and treatment.

▼ **Figure 49.24 Effects of addictive drugs on the reward system of the mammalian brain.** Addictive drugs alter the transmission of signals in the pathway formed by neurons of the ventral tegmental area (VTA), a region located near the base of the brain.



MAKE CONNECTIONS Review depolarization in Concept 48.3. What effect would you expect if you depolarized the neurons in the VTA? Explain.

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Alzheimer's Disease

The condition now known as **Alzheimer's disease** is a mental deterioration, or dementia, characterized by confusion and memory loss. Its incidence is age related, rising from about 10% at age 65 to about 35% at age 85. Overall, Alzheimer's disease accounts for about two of every three cases of dementia. It is also the sixth most common cause of death among adults in the United States, affecting individuals including former President Ronald Reagan, author E.B. White, and civil rights heroine Rosa Parks.

Alzheimer's disease is progressive; patients gradually become less able to function and eventually need to be dressed, bathed, and fed by others. Individuals with Alzheimer's disease often lose their ability to recognize people and may treat even immediate family members with suspicion and hostility.

Examining the brains of individuals who have died of Alzheimer's disease reveals two characteristic